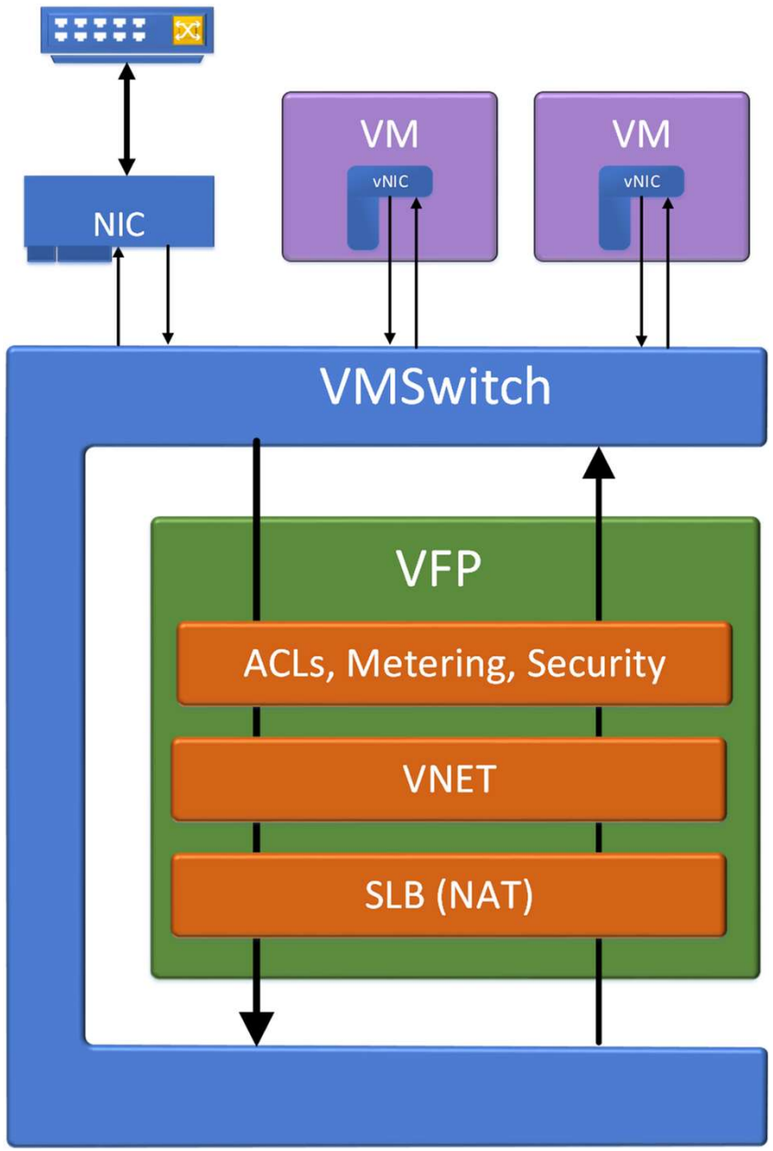
Azure Accelerated Networking: SmartNICs in the Public Cloud

Daniel Firestone, Andrew Putnam, Sambhrama Mundkur, Derek Chiou, Alireza Dabagh, Mike Andrewartha, Hari Angepat, Vivek Bhanu, Adrian Caulfield, Eric Chung, Harish Kumar Chandrappa, Somesh Chaturmohta, Matt Humphrey, Jack Lavier, Norman Lam, Fengfen Liu, Kalin Ovtcharov, Jitu Padhye, Gautham Popuri, Shachar Raindel, Tejas Sapre, Mark Shaw, Gabriel Silva, Madhan Sivakumar, Nisheeth Srivastava, Anshuman Verma, Qasim Zuhair, Deepak Bansal, Doug Burger, Kushagra Vaid, David A. Maltz, and Albert Greenberg

Daniel Firestone

Tech Lead and Group Manager, Azure Host Networking Team





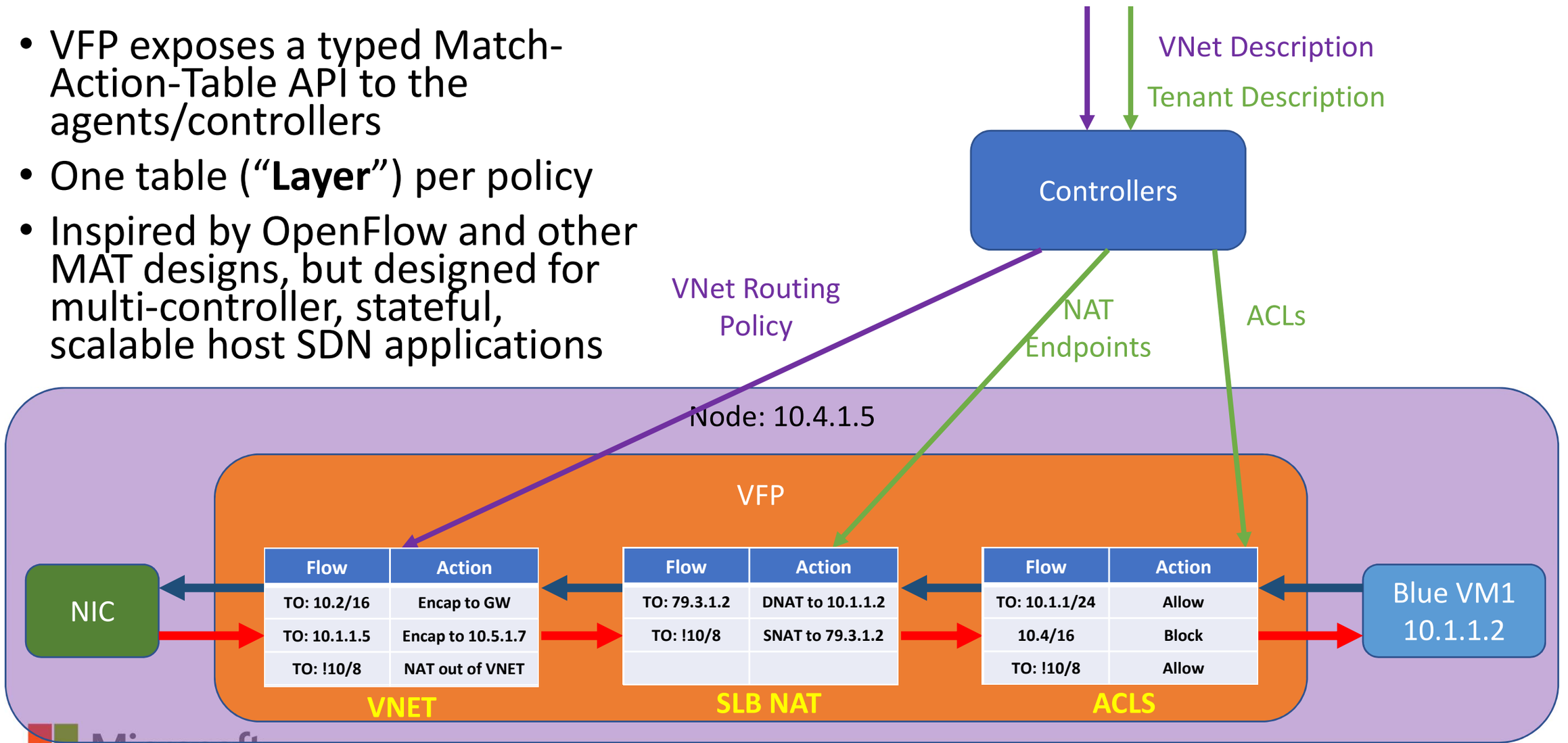
Virtual Filtering Platform (VFP) Azure's SDN Dataplane

- Virtual switch for Hyper-V / Azure
- Provides core SDN functionality for Azure networking services, including:
  - Address Virtualization for VNET
  - VIP -> DIP Translation for SLB
  - ACLs, Metering, and Security Guards
- Uses programmable rule/flow tables to perform per-packet actions
- Programmed by multiple Azure SDN controllers, supports all dataplane policy at line rate with offloads

Microsoft

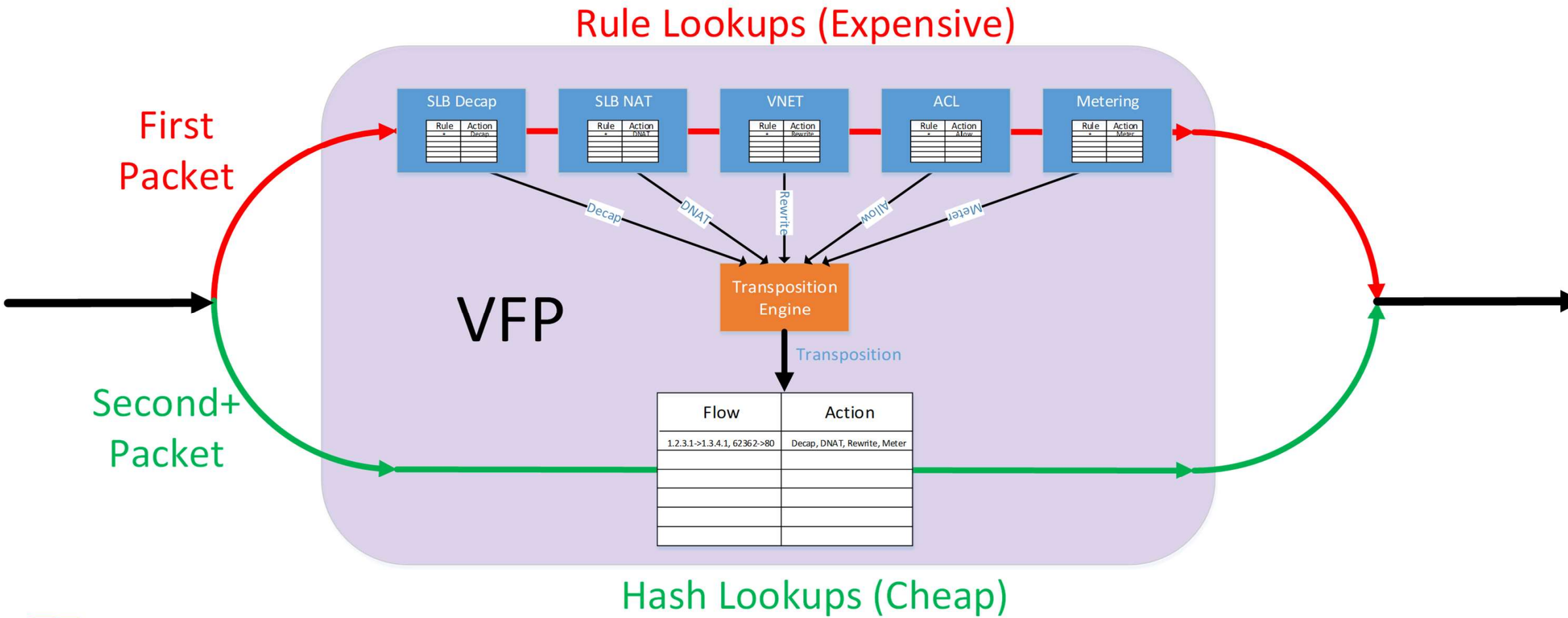
Key Primitive: Match Action Tables

- VFP exposes a typed Match-Action-Table API to the agents/controllers
- One table ("**Layer**") per policy
- Inspired by OpenFlow and other MAT designs, but designed for multi-controller, stateful, scalable host SDN applications



Microsoft

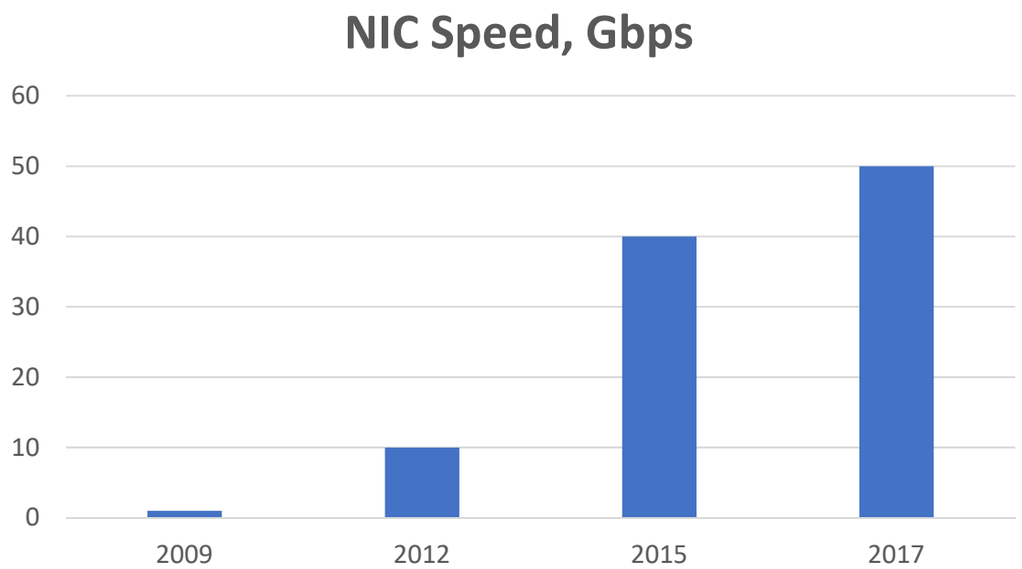
Unified Flow Tables – A Fastpath Through VFP



Microsoft

Scaling Up SDN: NIC Speeds in Azure

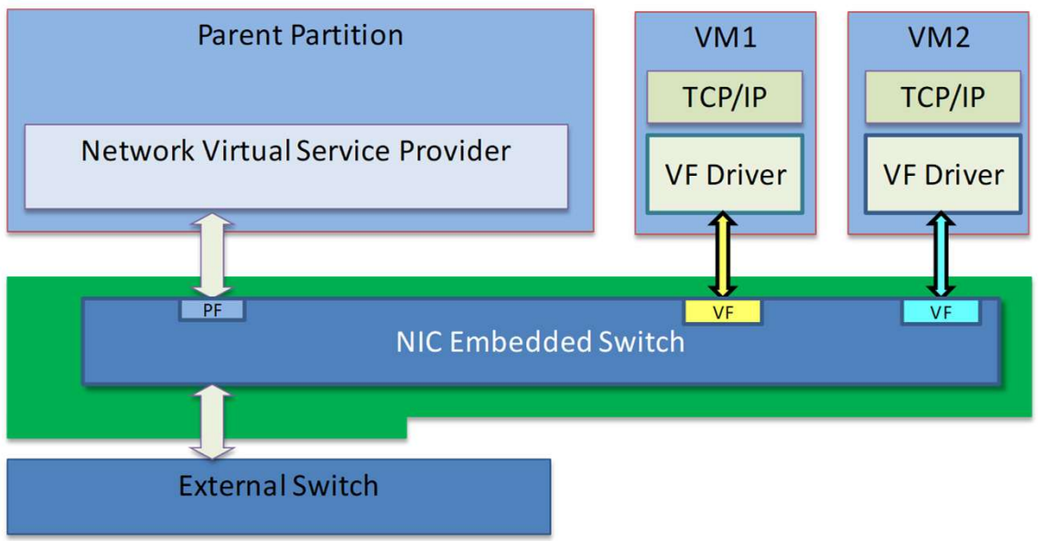
- 2009: 1Gbps
- 2012: 10Gbps
- 2015: 40Gbps
- 2017: 50Gbps
- Soon: 100Gbps?



We got a 50x improvement in network throughput, but not a 50x improvement in CPU power!

Microsoft

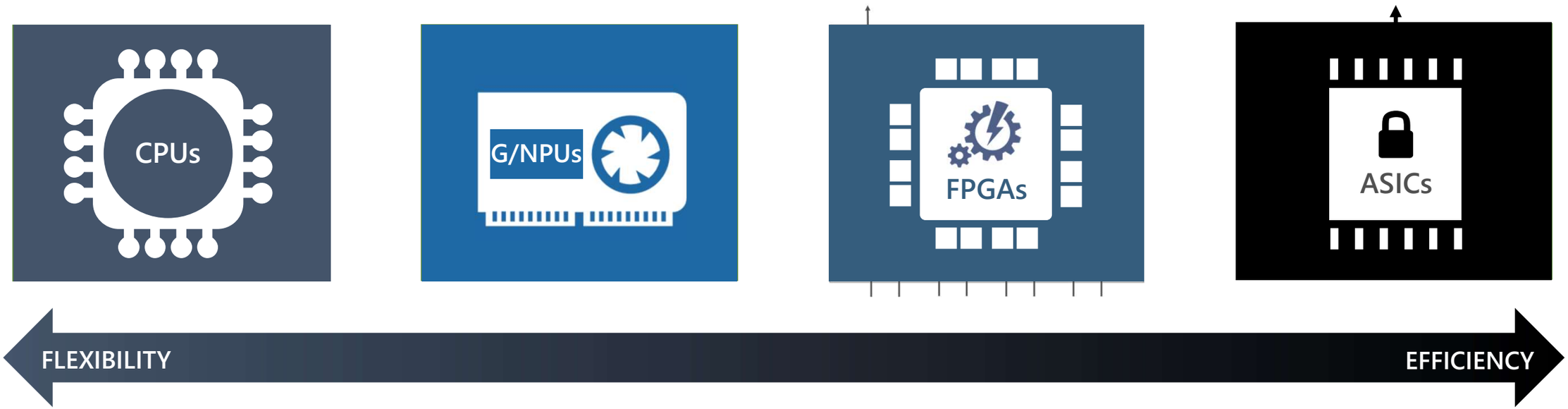
Example ASIC Solution: Single Root IO Virtualization (SR-IOV) gives native performance for virtualized workloads



But where is the SDN Policy?

Microsoft

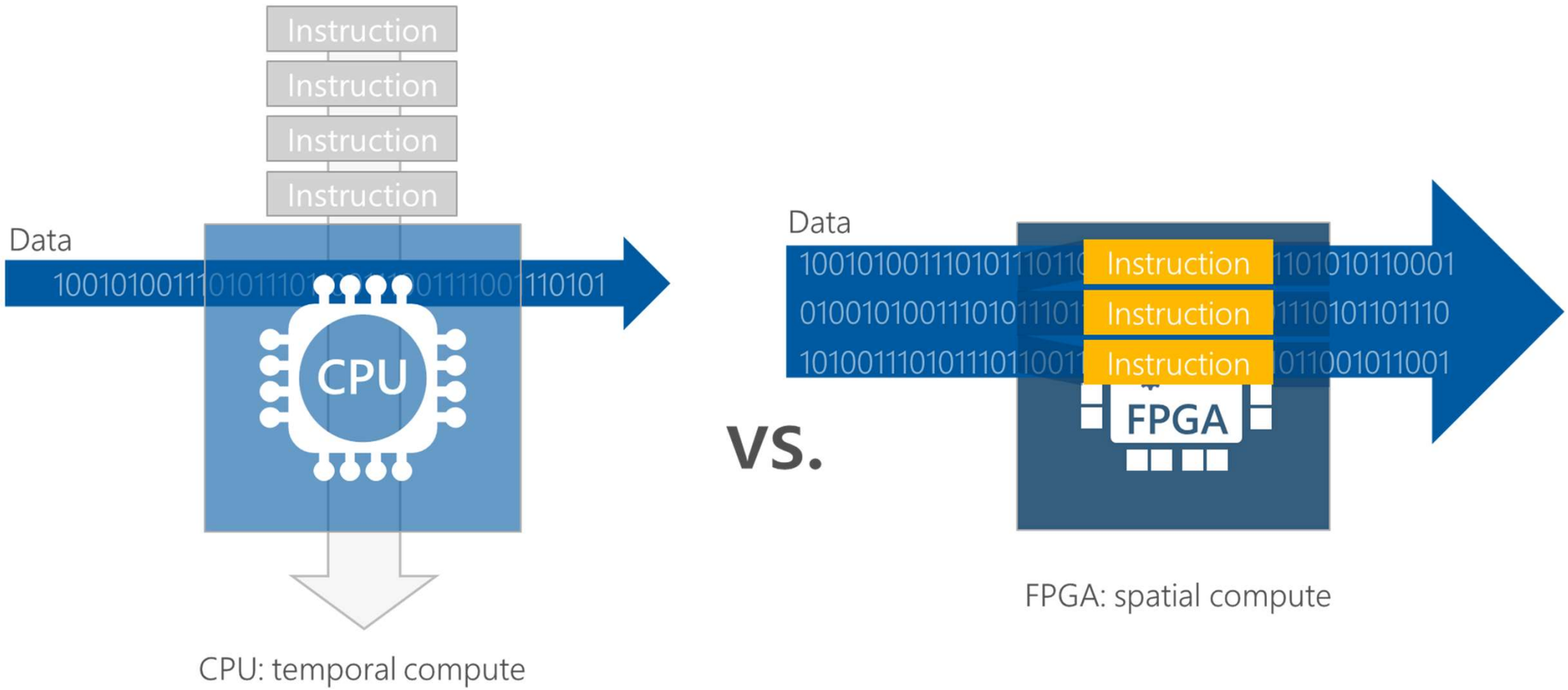
Silicon alternatives



Option 5: Don’t offload at all, instead make SDN more efficient with e.g. poll-mode DPDK

Microsoft

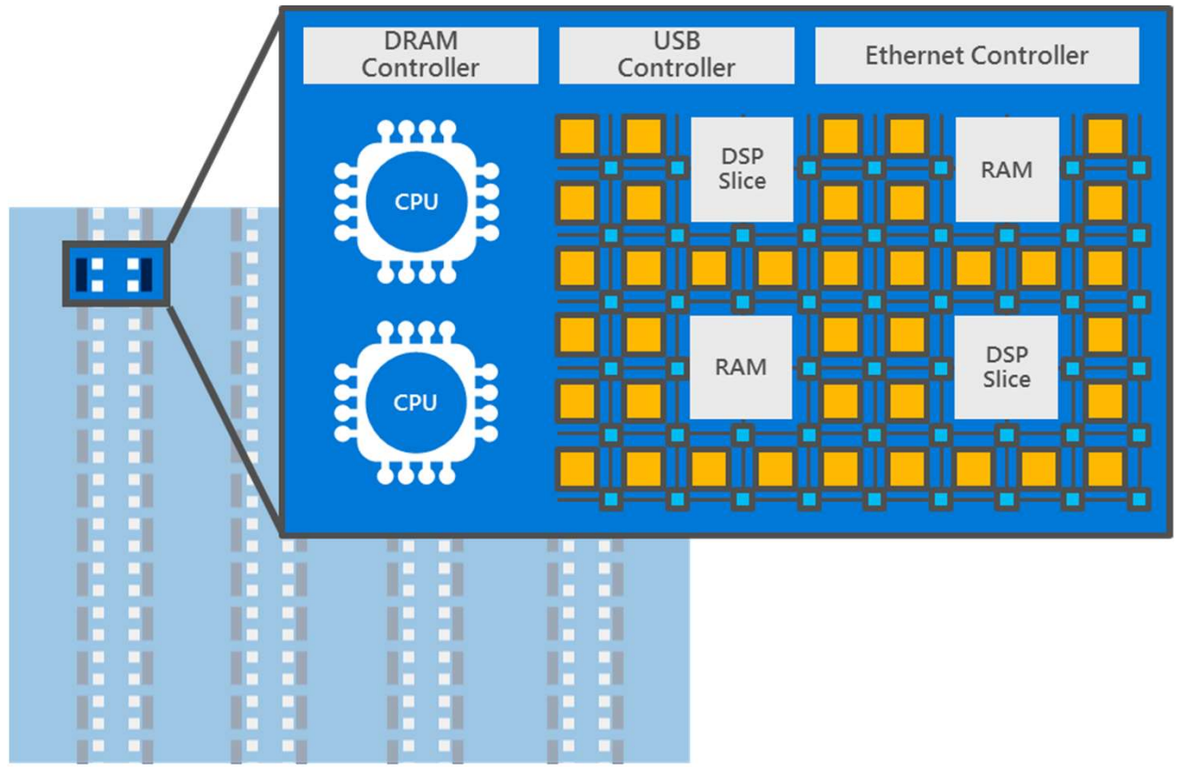
CPU vs. FPGA



Microsoft

What is an FPGA, Really?

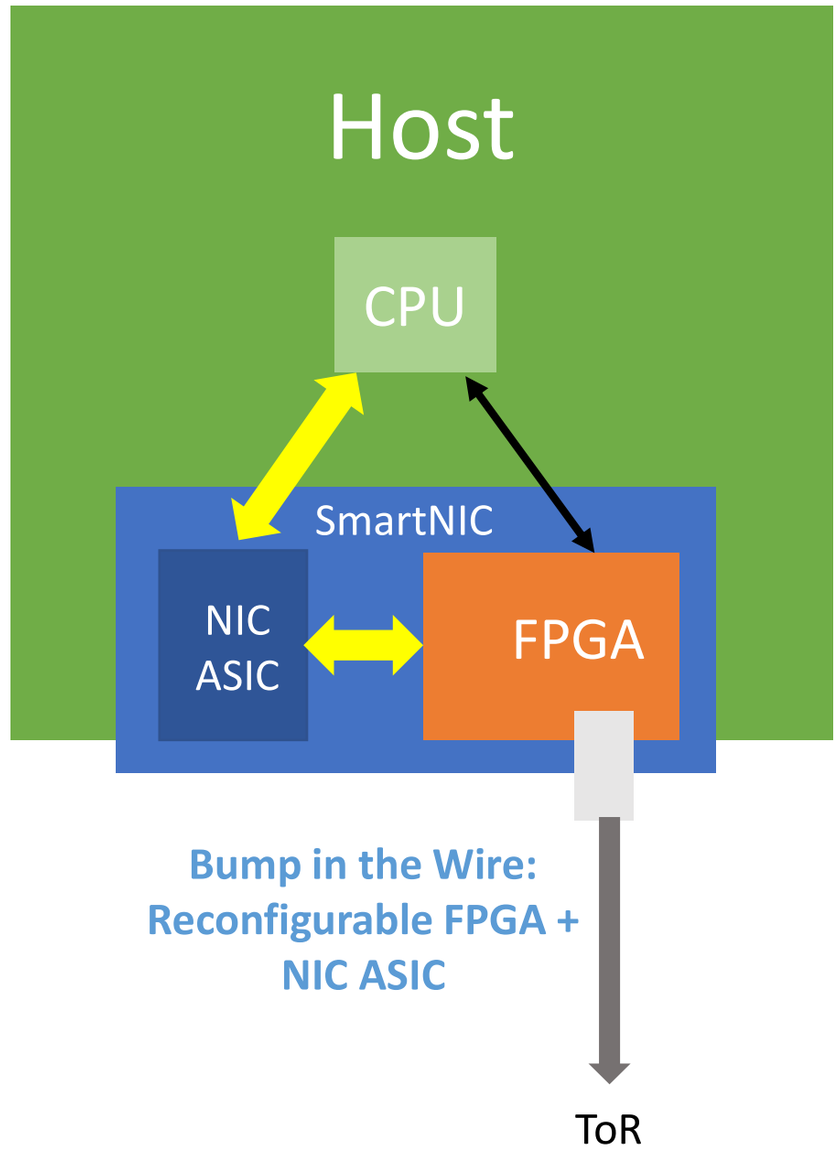
- Field Programmable Gate Array
- Chip has large quantities of programmable gates – highly parallel
- Program specialized circuits that communicate directly
- Two kinds of parallelism:
  - Thread-level parallelism (stamp out multiple pipelines)
  - Pipeline parallelism (create one long pipeline storing many packets at different stages)
- FPGA chips are now large SoCs (can run a control plane)



Microsoft

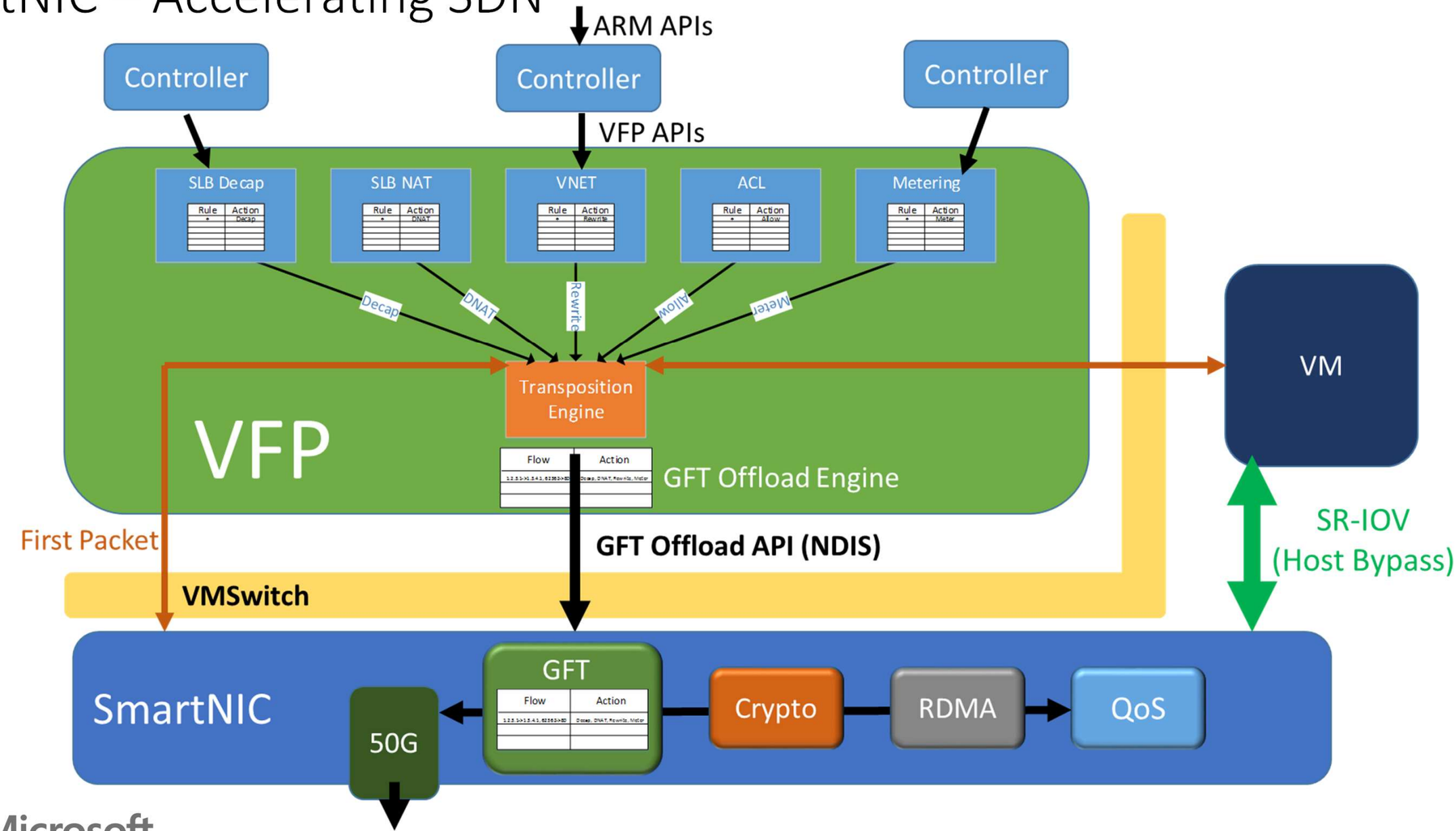
Our Solution:
Azure SmartNIC (FPGA)

- HW is needed for scale, perf, and COGS at 40G+
- 12-18 month ASIC cycle + time to roll new HW is too slow
- To compete and react to new needs, we need agility – SDN
- Programmed using Generic Flow Tables
  - Language for programming SDN to hardware
  - Uses connections and structured actions as primitives



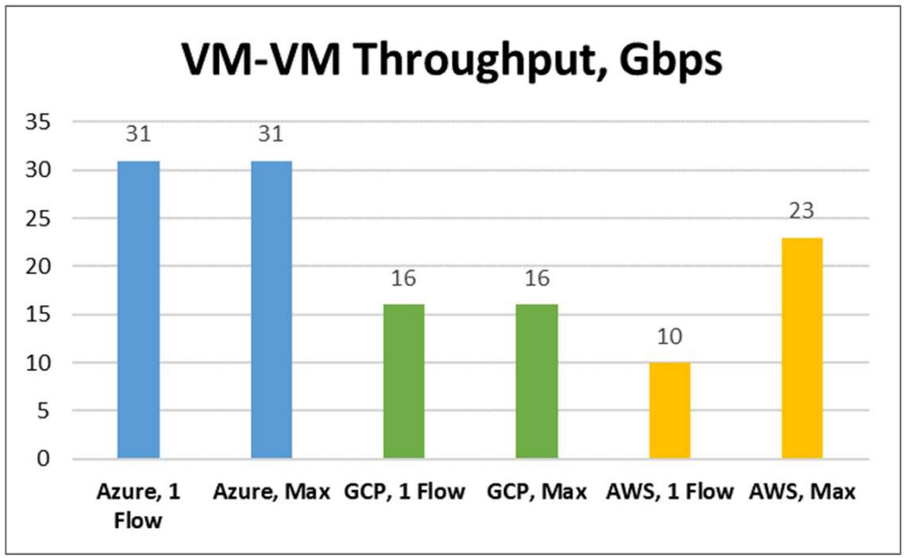
Microsoft

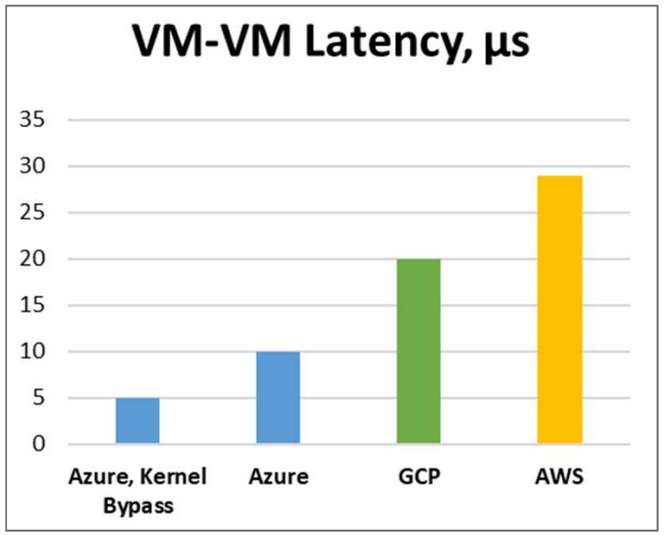
SmartNIC – Accelerating SDN

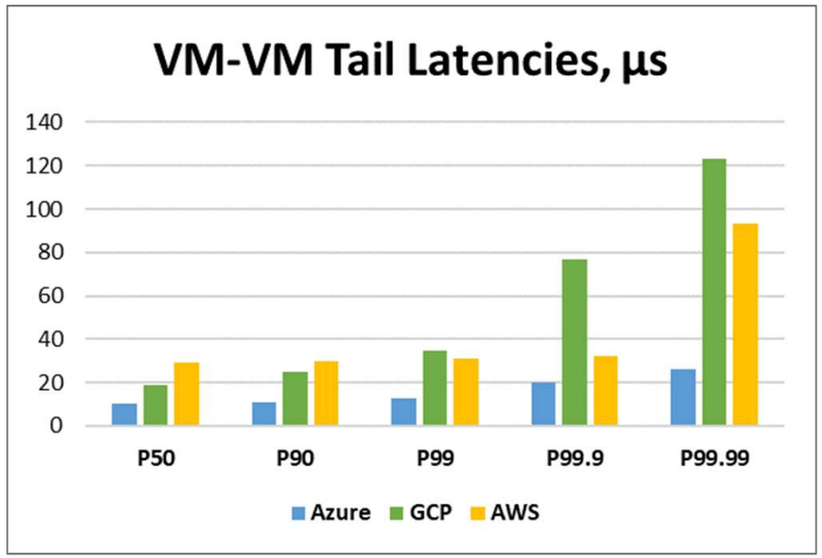


Microsoft

AccelNet Comparative Results







Microsoft

Conclusion:
Software or Hardware?

Each has its place…

Microsoft